

YouTube → Shorts

Complete Website Building Roadmap

For Student Developers | Vibe Coding Edition

Features • Security • Monetization • Tech Stack

■ 0

Startup Cost

8 Weeks

To Launch

■ 25K+

Month 3 Potential

100%

Free Tools

What You Are Building

An AI-powered web app that lets YouTubers and content creators paste a YouTube video URL and automatically get multiple short clips (Reels, TikToks, YouTube Shorts) with captions, titles, and hashtags — saving hours of editing work.

■ YouTube URL Input

Paste any YouTube link, app downloads the video

>■ AI Smart Clipping

AI finds the most engaging moments automatically

■ Auto Captions

Whisper AI adds animated subtitles to every clip

■ 9:16 Resize

Auto-converts to Reels/TikTok/Shorts format

■■ AI Title + Tags

GPT generates titles and hashtags for each clip

■■ Bulk Download

Download all clips in one ZIP file

■ Background Music

Royalty-free music library for clips

■ Auto-Schedule

Post directly to Instagram, TikTok (Pro plan)

Free Tech Stack (■0 Cost)

Layer	Tool	Why Free?	What It Does
Frontend	Next.js + Tailwind	Open source	Website UI & pages
Hosting	Vercel	Free tier forever	Deploy in 1 click
Backend	FastAPI (Python)	Open source	Handles video processing
AI Transcribe	Whisper AI	Open source by OpenAI	Auto captions
Video Edit	FFmpeg	Free forever	Cut & resize videos
YouTube DL	yt-dlp	Open source	Download YouTube videos
Database	Supabase	Free 500MB tier	User data & files
Storage	Cloudflare R2	Free 10GB/month	Store processed clips
Auth/Login	Clerk / NextAuth	Free tier	Secure user login
Payments	Razorpay	Free to integrate	Collect subscription fees
Security	Cloudflare WAF	Free plan	Block hackers & bots
Analytics	Posthog	Free 1M events	Track user behaviour
Error track	Sentry	Free 5K errors	Catch bugs instantly

GitHub Student Pack gives you ■50,000+ worth of free dev tools — sign up at education.github.com with your college email!

Week-by-Week Roadmap

Week 1–2	Setup & UI	Install Next.js, build landing page, set up Vercel deploy, add Clerk login, create dashboard UI with Tailwind.
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Week 3–4	Core AI Engine	Set up FastAPI backend, integrate yt-dlp for YouTube download, add FFmpeg video cutting, connect Whisper for transcription.
Week 5–6	Smart Features	Add 9:16 auto-resize, animated captions, GPT-4 title/hashtag generation, music picker, bulk ZIP download.
Week 7–8	Security & Launch	Add Cloudflare WAF, rate limiting, input validation, Razorpay payments, error logging. Launch free beta to 100 users.
Month 3+	Scale & Earn	Add social scheduling, team workspaces, analytics, API access, mobile app. Start charging paid plans.

Full Feature List

Free Plan Features

>■ Auto smart clipping AI picks best 3 clips per video	■ Basic captions Whisper AI auto-transcription
■ 9:16 format Auto resize for Shorts/Reels	■■ Download clips Up to 2 videos per month
■■ AI title generator 5 title suggestions per clip	■ Basic dashboard View all your clips

Pro Plan — ■499/month

■■ Unlimited videos No monthly cap on processing	■ Custom branding Add your logo & watermark
■ Multi-language captions 10+ language support	■ Social scheduling Auto-post to Instagram/TikTok
■ Music library 500+ royalty-free tracks	■ Analytics dashboard Views, CTR, best clip stats

Creator Plan — ■999/month

■ Team workspaces Add 5 team members	■ API access Integrate into your own tools
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■ Priority processing

2x faster video processing

■ Viral score

AI predicts which clips go viral

■ Content calendar

Plan & schedule all content

■ White-label option

Resell under your own brand

Security — Protect Your Website from Hackers

Cloudflare WAF (Free)

Your first wall against hackers. Put your site behind Cloudflare — it automatically blocks DDoS attacks, SQL injection, XSS attacks, and bad bots BEFORE they even reach your server. Setup takes 10 minutes. FREE.

HTTPS / SSL Everywhere

Never run your site on plain HTTP. Vercel gives you free HTTPS automatically. All data between users and your server is encrypted. Hackers can't read it even if they intercept it.

Rate Limiting

Limit each user to 10 API requests per minute. This stops attackers from spamming your server, burning your API credits, or doing brute-force attacks. Use Upstash Redis (free) for this.

JWT Authentication Tokens

Never store passwords yourself. Use Clerk or NextAuth — they handle secure sessions with JWT tokens. Every API call must include a valid token, so anonymous attackers can't access your processing engine.

Input Validation (Zod / Pydantic)

Validate EVERY input from users before touching it. Use Zod in Next.js and Pydantic in FastAPI. This stops SQL injection, XSS, and malformed data from crashing your server.

Environment Variables

NEVER hardcode API keys in your code. Use .env files and add .env to .gitignore immediately. Put secrets in Vercel's environment variables panel. One leaked key = your entire API bill stolen.

File Type Validation

If you accept file uploads, always check the file type server-side — not just the extension. Only allow video formats. Reject executables, scripts, and PHP files. Use python-magic library.

Auth Middleware on Every Route

Every backend API route must check that the user is logged in and owns the resource they're accessing. Never trust the frontend — always verify server-side. A logged-out user should get 401 immediately.

Error Logging with Sentry (Free)

Install Sentry on day 1. It catches crashes and errors before hackers can exploit them. You get email alerts when something breaks. Free up to 5,000 errors/month.

Regular Dependency Updates

Run 'npm audit' and 'pip-audit' weekly. Hackers exploit outdated packages. Use Dependabot on GitHub to auto-create PRs for security updates. Takes 5 minutes to set up.

Security Launch Checklist

■ Cloudflare set up and DNS pointed correctly

■ HTTPS forced, HTTP redirects to HTTPS

■ All API keys in .env (not in code)

■ .env added to .gitignore
■ Rate limiting active on all API routes
■ Auth middleware on every protected route
■ File upload validation (type + size limit)
■ Input validation with Zod / Pydantic
■ Sentry installed and sending alerts
■ npm audit + pip-audit run, 0 critical issues

Monetization & Earnings

Plan	Price	Users/month limit	Key Features	Target Customer
Free	■0	2 videos	Clips, captions, download	Try before buy
Pro	■499/mo	Unlimited	All features, scheduling, music	Solo creators
Creator	■999/mo	Unlimited	Teams, API, white-label, priority	Agencies, big creators

Revenue Projections

Paying Users	Avg Plan	Monthly Revenue	Annual Revenue
50 users	■499	■24,950	■2,99,400
100 users	■499	■49,900	■5,98,800
200 users	■499	■99,800	■11,97,600
500 users	■600	■3,00,000	■36,00,000

Cursor AI Prompts for Vibe Coding

Copy-paste these prompts into Cursor AI or Claude to build each part of your app:

Landing Page
"Build a modern Next.js landing page for an AI video tool called ShortifyAI. Include a hero section with a URL input, features grid, pricing table with 3 tiers, and a waitlist form. Use Tailwind CSS. Make it mobile responsive."
YouTube Downloader
"Write a Python FastAPI endpoint that accepts a YouTube URL, validates it, downloads the video using yt-dlp to a temp folder, and returns the file path. Include error handling and rate limiting with slowapi."

AI Clip Generator
"Write Python code using FFmpeg and Whisper AI that: 1) transcribes a video, 2) finds the 5 most engaging 60-second segments based on transcript energy, 3) cuts those segments, 4) resizes to 9:16 format for Shorts. Use subprocess for FFmpeg."
Auto Captions
"Write code to add animated captions to a video using FFmpeg. Use Whisper AI for transcription, create .srt subtitle file, burn subtitles onto video with white text, black outline, centered at bottom. Return the captioned video file."
Razorpay Payments
"Integrate Razorpay subscription payments into a Next.js app. Create API routes for creating a subscription plan, handling webhooks, updating user plan in Supabase, and checking if user has active subscription on protected routes."
Security Middleware
"Add security middleware to my Next.js app: 1) Rate limit API routes to 10 req/min using Upstash Redis, 2) Validate all inputs with Zod, 3) Add auth check on protected routes using NextAuth session, 4) Add CORS headers. Show the middleware.ts file."

Launch Strategy (Student Smart Move)	
Step 1	Build landing page only (no backend yet) Get 100 email signups BEFORE building. Proves demand. Takes 2 days.
Step 2	Post in creator communities Share on Reddit r/NewTubers, Facebook Groups, LinkedIn. Tell them it's free in beta.
Step 3	Launch MVP to first 50 users Give them all Pro features free for 1 month in exchange for feedback.
Step 4	Fix top 3 complaints Talk to your users. Build what they actually want, not what you think they want.
Step 5	Start charging after 100 users Offer a 'founding member' price of ₹299/month (cheaper forever). 50 sign up = ₹15K/mo.
Step 6	Post content about building it Tweet/post your journey. 'I built this as a student' content goes viral and drives signups.

Competitor Analysis — Your Advantage		
Tool	Price/month	Your Advantage
Opus Clip	\$29 (~₹2400)	Yours is 5x cheaper, India-focused
Vidyo.ai	\$29 (~₹2400)	Target Hindi/regional language creators
Munch	\$49 (~₹4000)	Much cheaper, simpler for beginners

Pictory	\$23 (~₹1900)	Better shorts-specific features
Your App	₹499/month	Local pricing, Razorpay, Hindi UI option

Final Advice: You don't need money to start. You need consistency.

Spend Week 1 on the landing page only. Share it. Get 50 signups. Then build. Fifty paying users at ₹499 = ₹24,950/month — more than most entry-level jobs. And you built it as a student, for free.